

**SIMATS ENGINEERING**  
**ASSESSMENT TOOL 3: Short Answer Test**

**COURSE NAME:** Cloud Computing and  
Big Data Analytics

**COURSE CODE:** CSA1522

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**COURSE OUTCOME COVERED**

**CO1:** *Apply cloud computing technologies including hardware-software evolution, virtualization, web services, and service models to develop cloud-based solutions.* (BL2)

Assessment Tool Weightage: 40%

Dave's Taxonomy: Imitation (Level 1)  
Question Count: 5

**Total Marks: 50**

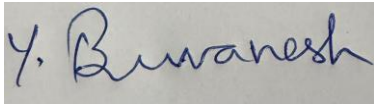
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**Code of Conduct**

I, Buvanesh Y (192372367) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



**Assessment Tasks:**

**Short Answer Test**

1. Define cloud computing and explain any four characteristics that make it different from traditional on-premise computing.
2. Trace the evolution of cloud computing from hardware evolution to internet software evolution, and show how these changes enabled modern cloud platforms.
3. Explain the role of server virtualization in cloud computing. Differentiate between virtualization as an enabling technology and cloud computing as a service model.
4. Compare SOAP-based and REST-based web services in terms of communication style, data exchange, and suitability for cloud applications.
5. Differentiate IaaS, PaaS, SaaS, and XaaS with one suitable real-world example for each model.

### Short Answer Test Rubric

| Criteria                     | Weightage | Level 4 - Exemplary                                           | Level 3 - Proficient                          | Level 2 - Developing                         | Level 1 - Beginning                 |
|------------------------------|-----------|---------------------------------------------------------------|-----------------------------------------------|----------------------------------------------|-------------------------------------|
| Conceptual Accuracy          | 30%       | Accurate definitions and precise explanation of all key terms | Mostly accurate with minor gaps               | Partial understanding with noticeable errors | Incorrect or irrelevant explanation |
| Coverage of Concepts         | 20%       | Covers all expected sub-points completely                     | Covers most expected points                   | Covers only basic points                     | Very limited coverage               |
| Use of Examples              | 15%       | Uses highly relevant cloud examples to strengthen the answer  | Uses relevant examples with minor mismatch    | Uses vague or partially relevant examples    | No useful examples                  |
| Comparison / Differentiation | 20%       | Clearly distinguishes concepts with strong logic              | Reasonable differentiation with minor overlap | Comparison is weak or incomplete             | Fails to differentiate concepts     |
| Clarity of Expression        | 15%       | Clear, organized, and concise answer writing                  | Generally clear with few issues               | Understandable but poorly organized          | Difficult to follow                 |

## 1. Cloud Computing: Definition and Four Key Characteristics

**Definition:** Cloud computing is the on-demand delivery of computing resources — servers, storage, databases, networking, software, and analytics — over the internet ("the cloud") on a pay-as-you-go basis, eliminating the need for users to own or manage physical infrastructure.

### Four distinguishing characteristics vs. traditional on-premise:

**On-demand self-service:** Users can provision resources (e.g., spin up a VM or increase storage) instantly through a portal without human involvement from the provider. On-premise requires raising a purchase order, waiting for hardware delivery, and manual setup — often weeks.

**Broad network access:** Cloud services are accessible over standard networks from any device (laptop, phone, tablet). On-premise resources are typically locked to a local network or require complex VPN setups for remote access.

**Resource pooling (multi-tenancy):** A provider's infrastructure is shared across many customers simultaneously, with resources dynamically assigned based on demand. On-premise dedicates hardware to a single organization, leading to idle capacity.

**Rapid elasticity:** Resources scale up or down automatically in minutes to match workload. A traditional data center must over-provision hardware upfront to handle peak load, wasting capacity the rest of the time.

**Measured service (bonus):** Cloud usage is metered and billed only for what is consumed, replacing large capital expenditure (CapEx) with predictable operational expenditure (OpEx).

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## 2. Evolution of Cloud Computing

### Hardware evolution → Virtualization era (1960s–2000s)

- Mainframes in the 1960s introduced the concept of *time-sharing* — multiple users sharing a single powerful machine. This is the philosophical ancestor of cloud.
- The PC revolution (1980s) shifted computing to on-premise desktops, but also drove down hardware costs dramatically.
- In the 1990s, x86 commodity servers became powerful enough to run enterprise workloads cheaply. Data centers filled with racks of physical servers, but utilization rates were poor (often 5–15%).
- VMware (1999) commercialized *server virtualization*, allowing one physical machine to run many isolated virtual machines. This dramatically improved utilization and is the direct technical foundation of IaaS.

### Internet/software evolution (1990s–2000s)

- The web (1991–1995) established HTTP as a universal, platform-agnostic communication protocol.
- Application Service Providers (ASPs) in the late 1990s delivered software over the internet — an early precursor to SaaS — but lacked the scalability and self-service that define modern cloud.
- Web services (SOAP, then REST) standardized how software components communicate over the internet, enabling loosely coupled, distributed architectures.

- Broadband proliferation made high-speed internet access ubiquitous, removing the bandwidth bottleneck that had limited earlier attempts.

### Convergence → Modern cloud platforms (2006–present)

- Amazon Web Services (2006) combined commodity hardware, virtualization, and internet delivery into the first true public IaaS platform (EC2, S3).
- Google App Engine (2008) and Microsoft Azure (2010) followed, adding PaaS layers.
- The result: any organization could consume enterprise-grade computing on demand, paying per hour, with no upfront capital investment.

### 3. Server Virtualization vs. Cloud Computing

#### Role of server virtualization in cloud:

Virtualization allows a single physical server's resources (CPU, RAM, disk) to be partitioned into multiple isolated *Virtual Machines (VMs)*, each running its own OS and workloads. In cloud computing, this is the engine that enables:

- **Multi-tenancy:** Many customers' VMs coexist on shared hardware without interfering with each other.
- **Rapid provisioning:** A new VM can be created in seconds from a template, rather than racking a new server.
- **Live migration:** VMs can be moved between physical hosts for maintenance or load balancing, transparently to the user.
- **Elasticity:** Resources can be allocated/deallocated dynamically by adjusting VM configurations.

#### Differentiation — Virtualization as enabling technology vs. Cloud as a service model:

| Dimension    | Virtualization                              | Cloud Computing                         |
|--------------|---------------------------------------------|-----------------------------------------|
| What it is   | A <i>technology</i> that abstracts hardware | A <i>service delivery model</i>         |
| Who uses it  | System administrators, IT ops               | End users, developers, businesses       |
| Scope        | Runs on a single physical server or cluster | Spans globally distributed data centers |
| Self-service | Requires admin intervention                 | Users provision resources themselves    |
| Billing      | Internal cost allocation                    | Pay-per-use metered billing             |
| Example      | VMware vSphere in your own data center      | AWS EC2                                 |

**In short:** You can have virtualization without cloud (a company running VMs internally), but you cannot have IaaS cloud without virtualization (or a modern equivalent like containerization). Virtualization is a *necessary but not sufficient* condition for cloud computing.

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## 4. SOAP vs. REST Web Services

| Aspect              | SOAP                                                                  | REST                                                                   |
|---------------------|-----------------------------------------------------------------------|------------------------------------------------------------------------|
| Full name           | Simple Object Access Protocol                                         | Representational State Transfer                                        |
| Communication style | Strictly RPC (Remote Procedure Call) via XML envelopes                | Resource-oriented; uses standard HTTP methods (GET, POST, PUT, DELETE) |
| Data format         | XML only                                                              | JSON (typically), XML, YAML, plain text — format-agnostic              |
| Protocol            | Can run over HTTP, SMTP, TCP, etc.                                    | HTTP/HTTPS only                                                        |
| Contract            | Requires a formal WSDL (Web Services Description Language) contract   | No mandatory contract; often documented with OpenAPI/Swagger           |
| State               | Stateless or stateful                                                 | Stateless by design                                                    |
| Overhead            | High — XML parsing and envelope wrapping add payload size and latency | Low — lightweight, especially with JSON                                |
| Error handling      | Built-in standardized fault messages                                  | Relies on HTTP status codes (200, 404, 500, etc.)                      |
| Security            | WS-Security standard built in                                         | Relies on HTTPS, OAuth, API keys                                       |

### Suitability for cloud applications:

REST is overwhelmingly preferred for cloud-native applications because its stateless, lightweight nature aligns with horizontal scaling (each request is independent, so any server can handle it). Its JSON payloads are smaller, reducing bandwidth costs. REST APIs are also simpler to consume from mobile and web frontends.

SOAP remains appropriate in enterprise contexts where strict contracts, transactions, and built-in security standards (e.g., WS-ReliableMessaging) are required — such as banking and healthcare legacy integrations.

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## 5. IaaS, PaaS, SaaS, and XaaS — Differentiated with Examples

### IaaS — Infrastructure as a Service

Delivers raw virtualized computing infrastructure: VMs, storage, networking. The user manages the OS, middleware, and applications; the provider manages the physical hardware.

*Example: Amazon EC2* — a developer rents a virtual server, installs Ubuntu, configures a web server, and deploys their application. They control the stack above the hypervisor.

### PaaS — Platform as a Service

Delivers a managed platform (OS + runtime + middleware) on which developers deploy their code without managing underlying infrastructure. Focus shifts entirely to application logic.

*Example: Google App Engine* — a developer uploads their Python or Java application; Google automatically handles server provisioning, OS patching, load balancing, and scaling. The developer never touches a VM.

### SaaS — Software as a Service

Delivers fully functional software applications over the internet. The user simply uses the application through a browser or client; no infrastructure or platform management is involved at all.

*Example: Microsoft 365 (Office Online)* — employees write documents and send email through a browser. Microsoft manages everything: infrastructure, platform, application updates, backups.

### XaaS — Everything as a Service

An umbrella term for the expanding spectrum of services delivered via the cloud beyond the classic three models. It captures emerging categories where virtually any function is offered on-demand.

| Sub-category                   | Example                                                                           |
|--------------------------------|-----------------------------------------------------------------------------------|
| DBaaS (Database as a Service)  | Amazon RDS — managed relational database, no server admin                         |
| FaaS (Function as a Service)   | AWS Lambda — run individual code functions triggered by events, pay per execution |
| STaaS (Storage as a Service)   | Dropbox — cloud file storage without managing hardware                            |
| SECaaS (Security as a Service) | Cloudflare — DDoS protection and WAF delivered as a cloud service                 |

XaaS reflects the principle that *any* layer of the IT stack can be abstracted, managed by a provider, and consumed on-demand — the logical endpoint of the cloud computing evolution.